

# Comparitive Performance Report for direct-reads-3-way-replica-balance-reads- off vs direct-reads-3-way-replica-balance-reads- on-run2

# Table of contents

- Comparison summary for direct-reads-3-way-replica-balance-reads-off vs direct-reads-3-way-replica-balance-reads-on-run2
- Response Curves
  - Sequential Read
  - Random Read
  - Random Read/Write
- Configuration yaml files
  - results

# Comparison summary for direct-reads-3-way-replica-balance-reads-off vs direct-reads-3-way-replica-balance-reads-on-run2

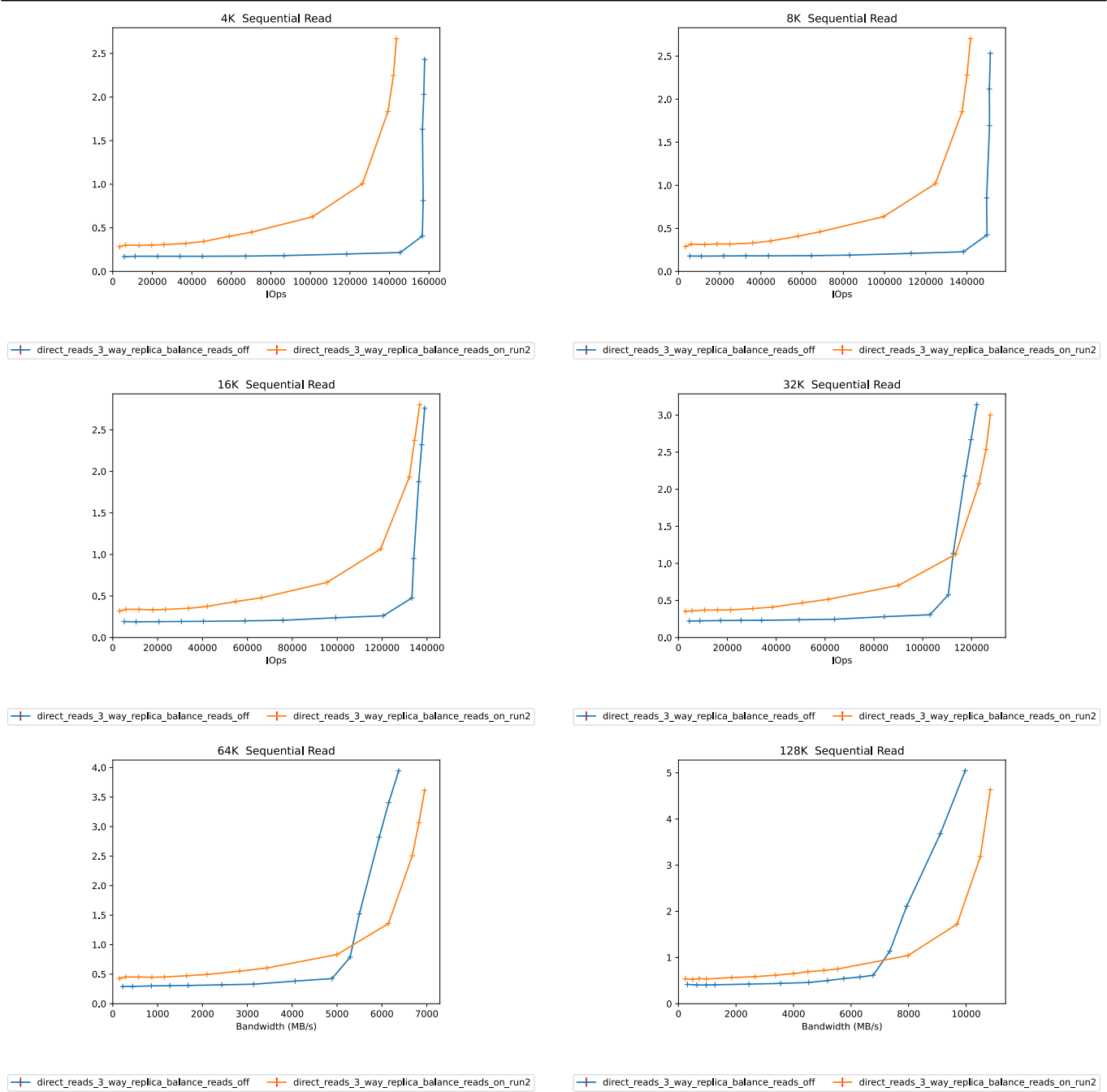
| Sequential Read | direct_reads_3_way_replica_balance3reads_off | direct_reads_3_way_replica_balance3reads_on | change throughput | change latency |
|-----------------|----------------------------------------------|---------------------------------------------|-------------------|----------------|
| 4K              | 157805 IOps@2.4ms                            | 143517 IOps@2.7ms                           | -9%               | 13%            |
| 8K              | 151340 IOps@2.5ms                            | 141720 IOps@2.7ms                           | -6%               | 8%             |
| 16K             | 138955 IOps@2.8ms                            | 136758 IOps@2.8ms                           | -2%               | 0%             |
| 32K             | 122155 IOps@3.1ms                            | 127663 IOps@3.0ms                           | 5%                | -3%            |
| 64K             | 6374 MB/s@3.9ms                              | 6952 MB/s@3.6ms                             | 9%                | -8%            |
| 128K            | 9968 MB/s@5.0ms                              | 10842 MB/s@4.6ms                            | 9%                | -8%            |
| 256K            | 14279 MB/s@7.0ms                             | 14710 MB/s@6.8ms                            | 3%                | -3%            |
| 384K            | 16025 MB/s@12.6ms                            | 15306 MB/s@13.1ms                           | -4%               | 4%             |
| 512K            | 17311 MB/s@15.5ms                            | 13513 MB/s@9.9ms                            | -22%              | -36%           |
| 768K            | 17046 MB/s@23.6ms                            | 13135 MB/s@9.6ms                            | -23%              | -59%           |
| 1024K           | 16582 MB/s@16.2ms                            | 13688 MB/s@4.9ms                            | -17%              | -70%           |
| 2048K           | 14981 MB/s@17.9ms                            | 14427 MB/s@9.3ms                            | -4%               | -48%           |
| 4096K           | 14188 MB/s@37.8ms                            | 14250 MB/s@18.8ms                           | 0%                | -50%           |

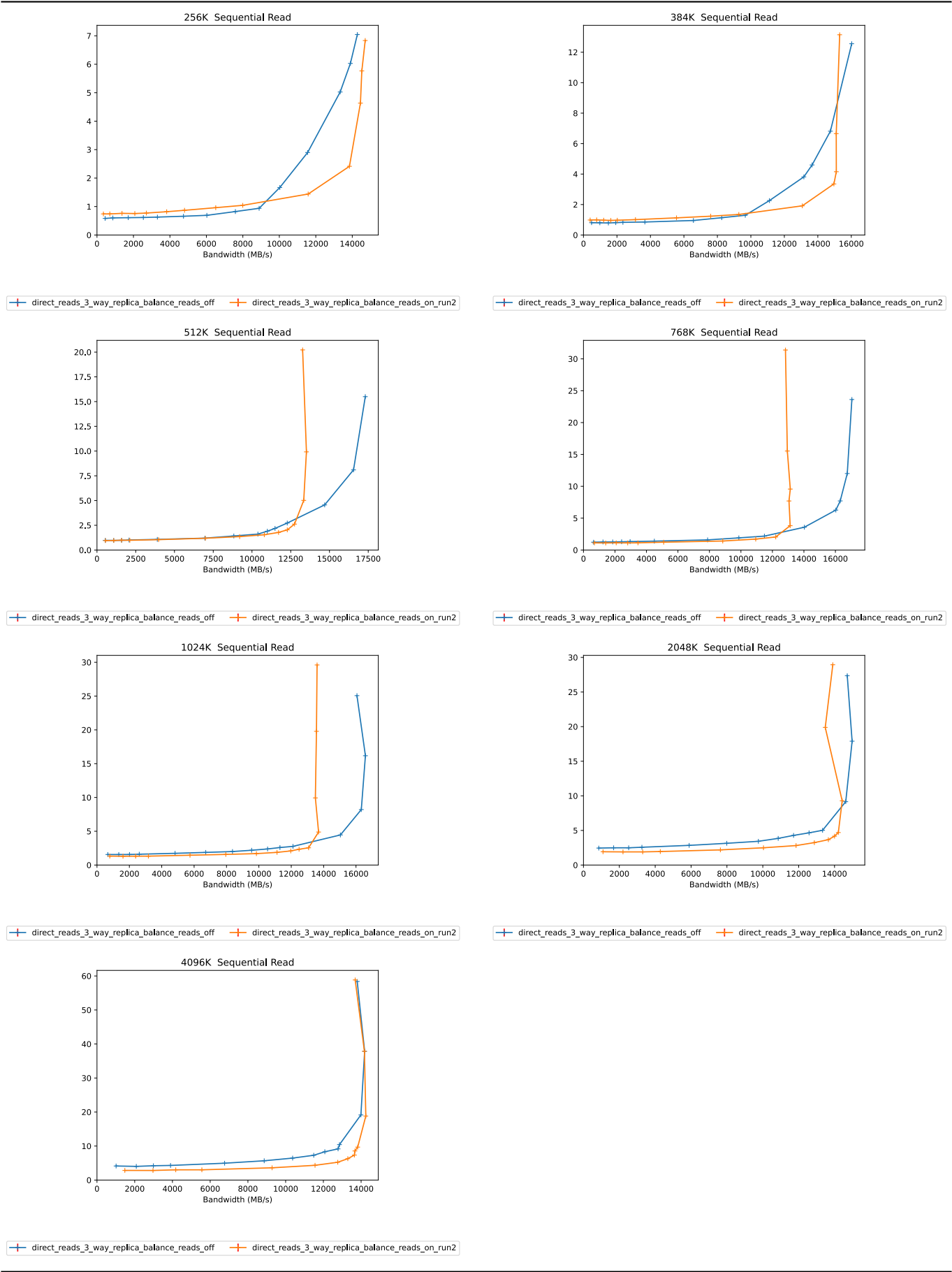
| Random Read | direct_reads_3_way_replica_balance3reads_off | direct_reads_3_way_replica_balance3reads_on | change throughput | change latency |
|-------------|----------------------------------------------|---------------------------------------------|-------------------|----------------|
| 4K          | 223682 IOps@1.1ms                            | 153022 IOps@2.5ms                           | -32%              | 127%           |
| 8K          | 216775 IOps@1.8ms                            | 150278 IOps@2.6ms                           | -31%              | 44%            |
| 16K         | 193155 IOps@1.7ms                            | 143622 IOps@2.7ms                           | -26%              | 59%            |
| 32K         | 167800 IOps@1.9ms                            | 133689 IOps@2.9ms                           | -20%              | 53%            |
| 64K         | 8939 MB/s@2.3ms                              | 7397 MB/s@3.4ms                             | -17%              | 48%            |
| 128K        | 13489 MB/s@2.5ms                             | 11004 MB/s@4.6ms                            | -18%              | 84%            |
| 256K        | 16720 MB/s@6.0ms                             | 13769 MB/s@4.9ms                            | -18%              | -18%           |
| 384K        | 17456 MB/s@5.8ms                             | 14159 MB/s@3.5ms                            | -19%              | -40%           |
| 512K        | 17682 MB/s@7.6ms                             | 12781 MB/s@5.2ms                            | -28%              | -32%           |
| 768K        | 17132 MB/s@23.5ms                            | 12701 MB/s@3.9ms                            | -26%              | -83%           |
| 1024K       | 16383 MB/s@16.4ms                            | 13345 MB/s@5.0ms                            | -19%              | -70%           |
| 2048K       | 14846 MB/s@18.1ms                            | 14732 MB/s@9.1ms                            | -1%               | -50%           |
| 4096K       | 13878 MB/s@58.0ms                            | 14455 MB/s@18.5ms                           | 4%                | -68%           |

| Random Read/Write | direct_reads_3_way_replica_balance3reads_off | direct_reads_3_way_replica_balance3reads_on | change throughput | change latency |
|-------------------|----------------------------------------------|---------------------------------------------|-------------------|----------------|
| 4K_70/30          | 122950 IOps@3.1ms                            | 96146 IOps@4.0ms                            | -22%              | 29%            |
| 16K_70/30         | 103921 IOps@3.7ms                            | 86022 IOps@4.5ms                            | -17%              | 22%            |
| 64K_30/70         | 3282 MB/s@7.6ms                              | 2794 MB/s@6.0ms                             | -15%              | -21%           |
| 64K_70/30         | 4961 MB/s@3.4ms                              | 4179 MB/s@6.0ms                             | -16%              | 76%            |

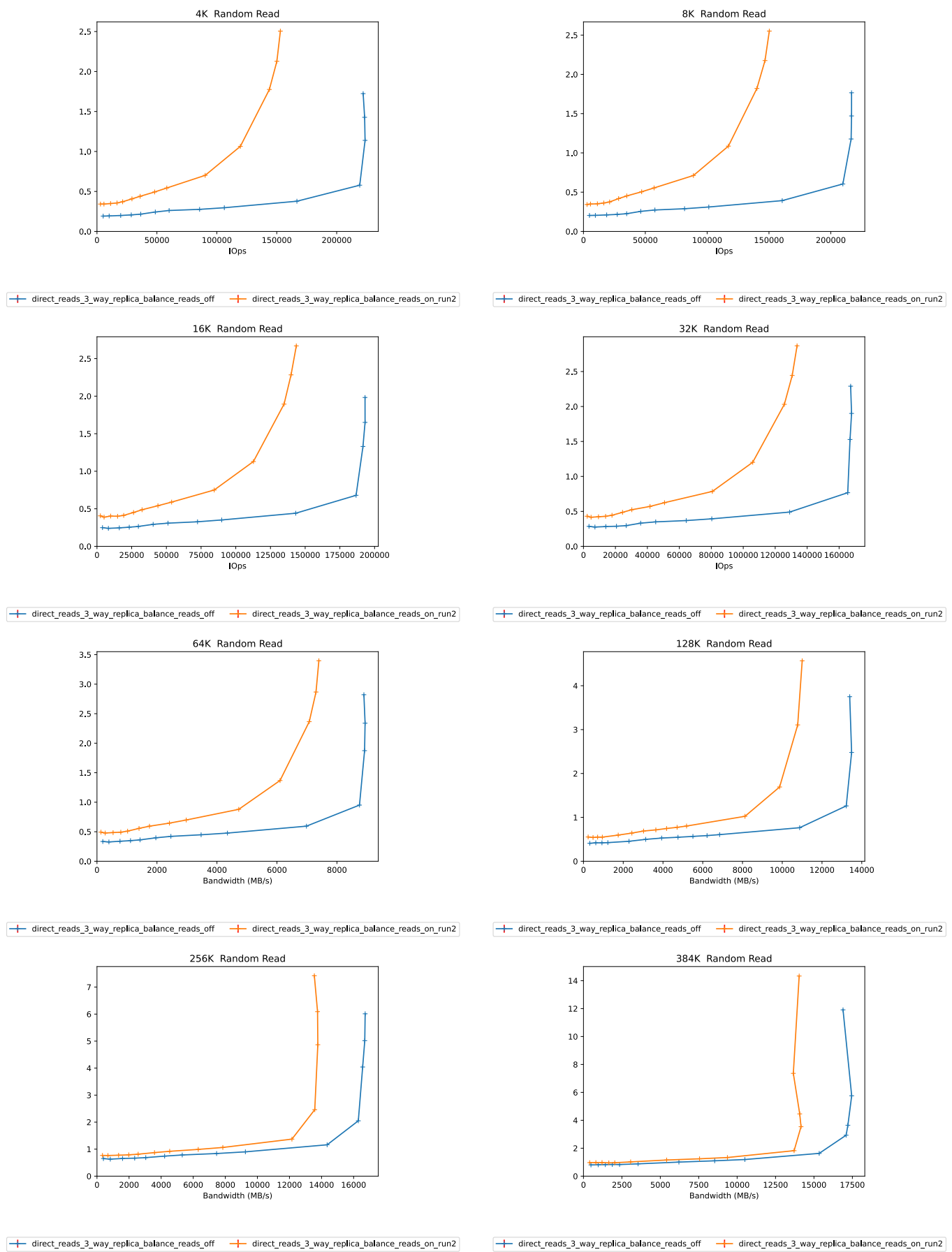
# Response Curves

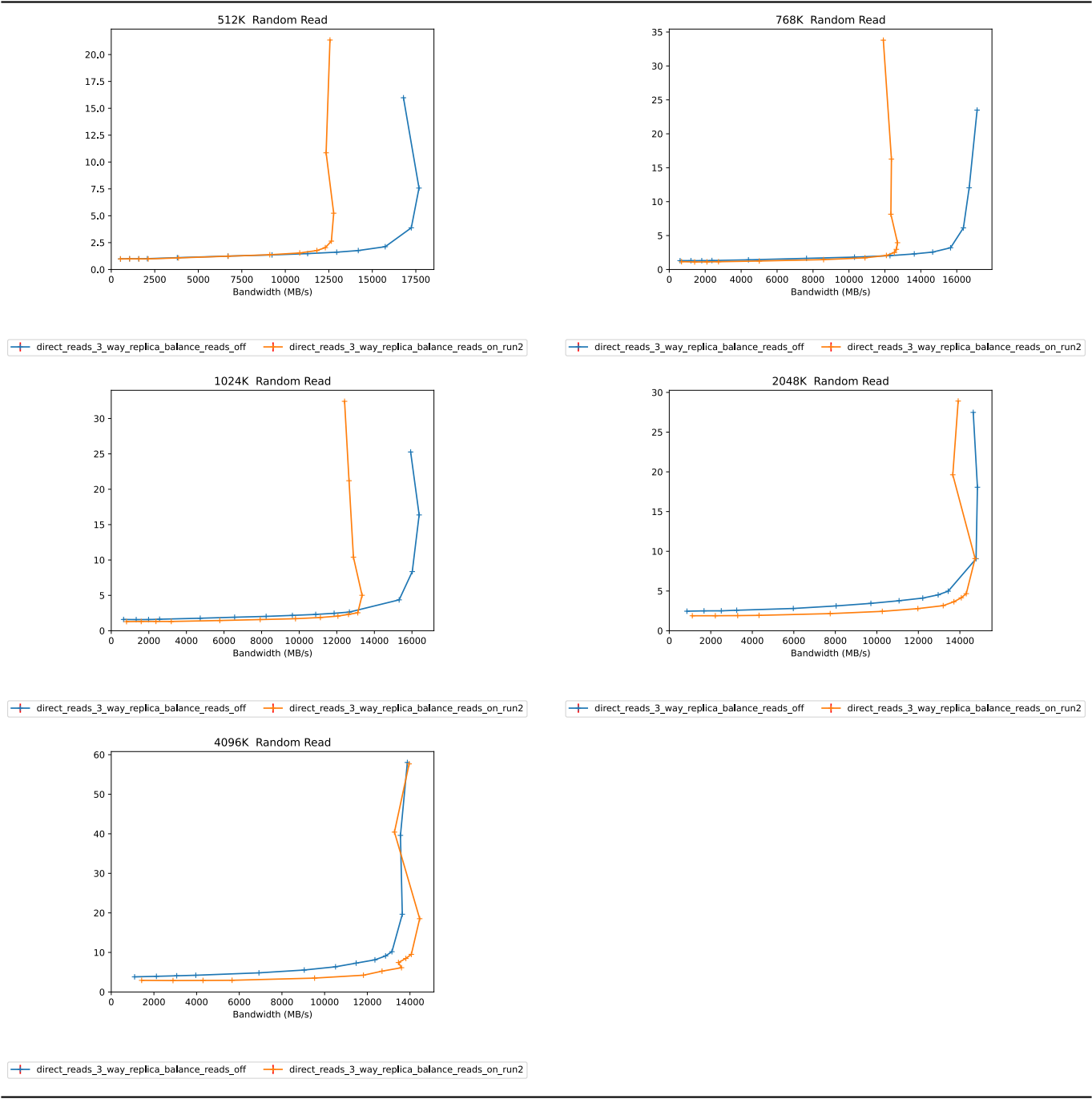
## Sequential Read



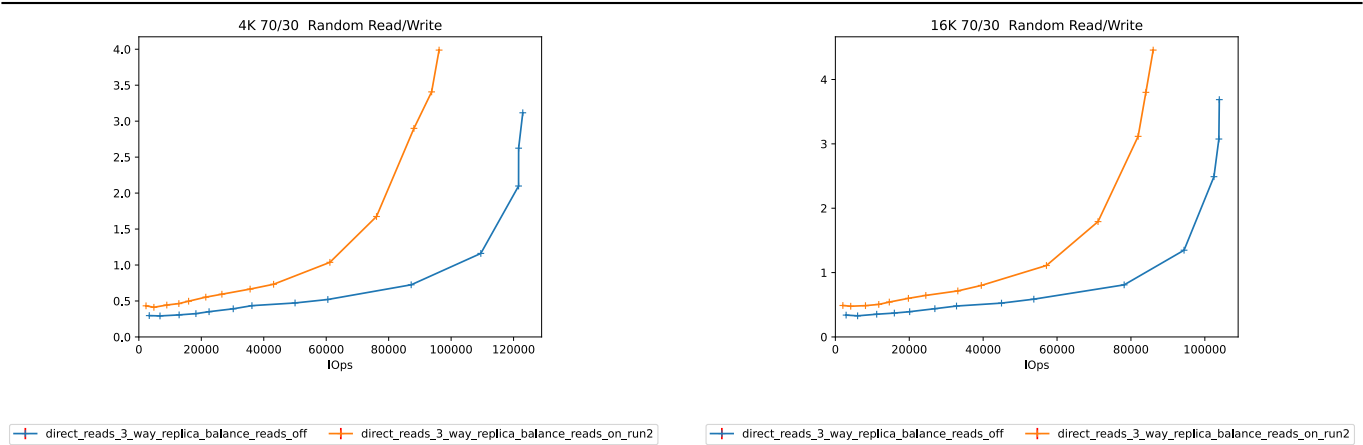


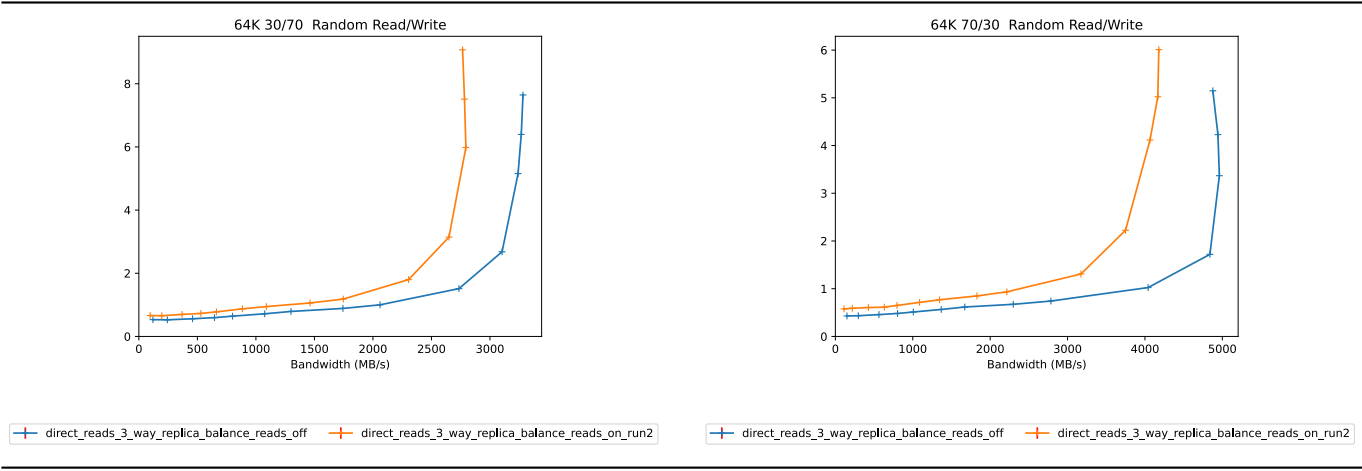
## Random Read





## Random Read/Write





# Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

## results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op\_size: 4096
- rwmixread: 70
- total\_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op\_size: 4096
- total\_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op\_size: 4096
- total\_iodepth:

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64kmixread70:
    jobname: mixread
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```